[20, 151, 151, 161]. Responsiveness can be quantified by the latency or sensitivity with which recommendation outputs reflect profile changes.

G5. Cultural / Linguistic Coverage. It evaluates whether the system surfaces music content from diverse cultures, languages, and regions, in accordance with the user's request or a fairness policy. It aims to ensure the LLM does not overly bias one language or culture [43, 87, 144]. Metrics such as cultural coverage or geo-diversity can be used to assess whether the recommendations fairly represent different regions within the global music catalog.

G6. Classical Relevance. It captures standard IR/RecSys performance using label-based metrics like Precision@k, Recall@k, and nDCG. While still informative, these should be interpreted in conjunction with grounding, personalization, and diversity metrics to reflect the multifaceted nature of music recommendation tasks.

We further discuss setting-specific bundles for *few-shot in-context learning (ICL)*, *retrieval-augmented generation (RAG)*, and *chain-/tree-of-thought (CoT/ToT)*.

S-ICL: Few-shot In-Context Learning. The evaluation of few-shot ICL might make use of several specialized metrics to assess the *stability* and *utility* of few-shot prompting in music recommendation. For instance, *shot calibration* [185] tests how performance changes as the number of examples (shots) varies; ideally, adding more examples should not drastically mislead the model. Zhang et al. [185] show that as we increase the number of in-context examples, models often become more miscalibrated before improving, with particularly poor calibration in low-shot regimes, highlighting the need to monitor this trade-off. The authors suggest measuring calibration curves across different shot counts (e.g., 0, 3, 5, 7) to ensure the model benefits from additional examples predictably. Zhou et al. [192] propose Batch Calibration to address the contextual bias that arises from varying numbers of in-context examples, showing that proper calibration techniques can recover performance degradation caused by prompt brittleness across different shot configurations.

Stability to prompt perturbations is another critical dimension: the model's recommendations should remain consistent under minor prompt modifications (e.g., reordering or paraphrasing the examples). For instance, Zhuo et al. [194] introduce PromptSensiScore (PSS), which quantifies prompt sensitivity as the average discrepancy in LLM responses when confronted with semantic variants of the same instruction. Their findings reveal that few-shot examples can alleviate sensitivity, with the most pronounced reduction occurring in the transition from zero-shot to one-shot settings. For measuring list-based recommendation stability, rank-biased overlap [168] provides a complementary assessment of ranking consistency under perturbations.

Additionally, *preference sensitivity* could be evaluated to ensure that when user preference cues (likes/dislikes) are given as examples, the recommender's output shifts appropriately along the intended facets (such as recommending more of a liked genre and fewer disliked styles). For instance, if a user's example indicates love of jazz and dislike of loud rock, the subsequent recommendations should reflect those biases measurably. Yu et al. [177] propose In-Context Preference Learning (ICPL), demonstrating that LLMs can learn reward functions from iteratively placed human feedback in context, achieving orders of magnitude greater sample efficiency than traditional preference-based methods. Their work shows that properly designed ICL can capture nuanced preference signals and translate them into consistent recommendation adjustments, validating that LLMs possess native preference-learning capabilities that enable few-shot adaptation to user tastes.

Finally, *instruction–example alignment* might check that the system continues to obey the explicit request instructions even when some in-context examples might be ambiguous or conflict with those instructions. In other words, the user's prompt requirements (e.g., "no mainstream hits") must dominate over any contradicting signals from the examples.

Zhou et al. [192] show that LLMs suffer from various bias factors in prompts, including formatting choices, verbalizer selection, and ICL examples, which can cause the model to deviate from instruction intent. Zhuo et al. [194] further show that instruction-following fidelity correlates with model confidence: higher decoding confidence (measured via token probabilities) corresponds to enhanced resilience against prompt variations and better instruction adherence.

Retrieval-Augmented Generation (S-RAG). The evaluation of an S-RAG pipeline spans multiple aspects across both the retrieval and recommendation stages. As noted by Es et al. [44], evaluating RAG systems requires assessing the retrieval module's ability to fetch relevant context and the generation module's ability to use that evidence faithfully. For the retrieval stage, one can use standard IR metrics (e.g., Recall@k of known relevant items) to gauge whether the LLM is retrieving useful evidence. At the recommendation generation stage, the primary focus is on *grounding*: conditioning the LLM on up-to-date catalog facts should reduce hallucinations and ensure that recommended tracks exist and meet the user's criteria. To quantify groundedness, one might define *EvidenceGrounding@k* as the proportion of the top-k recommended items whose key attributes can be justified by the retrieved items. This metric is inspired by faithfulness and citation-based evaluations proposed for RAG outputs [5, 45].

Another diagnostic is a document-swap test, denoted Δdoc, where we perturb the retrieved context (e.g., replacing a percentage of relevant items with irrelevant ones) to see if the model's recommendations change. A strong drop in recommendation quality after such a swap would indicate that the model truly relies on the retrieved evidence, whereas an invariant output would suggest that it ignores the provided items. This methodology is analogous to input-perturbation analyses used to study how LLMs utilize long contexts [91].

Beyond grounding, we also need to evaluate the *freshness*, *breadth*, and *efficiency* of RAG-based recommendations. For timeliness, *inspired by* work on time-aware evaluation and recency effects in recommender systems [14, 30], one might introduce freshness and temporal-bias metrics such as *Time-To-First-Surface (TTFS)* and *Day-0 Lift*, whose aim is to summarize how quickly new releases first appear in either the retrieved pool or the final top-k list, and how much extra exposure they receive on their release day. For example, a low TTFS and high Day-0 Lift would indicate that a newly released track already shows up quickly and prominently in users' recommendations instead of only weeks later.

To understand which parts of the catalogue the LLMs actually uses, we monitor simple coverage-style metrics *inspired by* beyond-accuracy evaluation [75, 148], notably metrics such as *UniqueArtists@k* counting distinct artists in the top-k recommendations, or *EntitiesResolvedShare* measuring the fraction of recommended items that can be matched to real catalogue entities (i.e., non-hallucinated tracks).

Finally, it is important to monitor the inference latency L (end-to-end response time), the total token budget B (number of tokens processed per query, including prompt, retrieved evidence, and generated output), and the evidence token ratio B_D/B (share of the budget devoted specifically to retrieved documents). An effective S-RAG configuration should improve freshness and catalogue coverage at both retrieval and recommendation stages, without incurring prohibitive cost, while still satisfying core recommendation goals such as accuracy, novelty, and personalization.

S-COT: Chain-of-Thought Reasoning. Complex reasoning for playlist curation requires both verifiable reasoning steps and coherent output. Here, one might use *Reasoning faithfulness*, which measures intermediate steps verifiable against catalogs [114]. In contrast, *reasoning impact* (ΔU_{CoT}) can be used to quantify the utility gain from explicit reasoning chains compared to direct answers [171]. Self-consistency uplift evaluates improvements from aggregating outputs across multiple reasoning paths [166]. Overall, one can state that the reasoning metrics integrate with query fidelity, diversity, and controllability dimensions.

The table below provides a concise summary and reference frame for our NL-based evaluation of LLM-driven MRS.

MRS-focused success evaluation dimensions

Core dimensions (G1–G6) — relevant across all LLM prompting settings:

Dimension	Examples of evaluation metrics
G1. Query Adherence & Groundedness	KB entity resolution, constraint satisfaction, hallucination penalty
G2. Discovery Quality	LongTailShare, Diversity@k, GeoDiv@k, genre/artist spread
G3. Personalization Gain	$\Delta U_{\text{profile}}$ (uplift with vs. without user context)
G4. Profile Fidelity & Controllability	Profile alignment, adaptation latency to explicit edits
G5. Cultural / Linguistic Coverage	CCR, language/region representation per request
G6. Classical Relevance	Precision@k, Recall@k, nDCG (read alongside G1–G5)

Setting-specific evaluation bundles — distinct per LLM prompting setting:

Setting	Examples of evaluation metrics and tests
S-Prompting — In-Context Learning & Chain-of-Thought	
ICL Stability	• Shot calibration curves (1, 3, 5, 10 examples) • PromptSensiScore, RBO for semantic/ranking consistency • Preference injection tests (genre, energy, tempo shifts) • Instruction dominance over noisy/conflicting examples
CoT Reasoning	• Reasoning faithfulness (step verifiability) • ΔU_{CoT} (chains shown vs. hidden) • Flow coherence (key, tempo, mood transitions) • Self-consistency uplift via multi-path aggregation
S-RAG — Retrieval-Augmented Generation	
Grounding	• EvidenceGrounding@k (output–passage justification) • Doc-swap test: Δ_{doc} with irrelevant retrieval • EntitiesResolvedShare, UniqueArtists@k
Freshness	• TTFS (Time-to-First-Surface), Day-0 Lift for new releases • Temporal bias detection (catalog recency distribution)
Efficiency	• Retrieval Recall@M, latency L, token cost B, B_D/B ratio

3.2.2 Measures of risks. Beyond success-oriented metrics, which capture how well a system fulfills user intent and provides relevant, grounded recommendations, it is equally important to account for potential *risks* and undesirable behaviors arising from generative models in music recommendation. For this reason, we include a complementary set of **diagnostic metrics and dimensions** addressing issues such as hallucination, exposure skew, privacy and

stereotyping, automatic-judge bias, and cross-lingual failure. These diagnostics provide a broader understanding of system robustness, fairness, and trustworthiness—dimensions that classical success metrics alone cannot capture.

1) **Hallucinations and entity errors**: This dimension measures instances of fabrication (generating non-existent entities), misattribution (assigning incorrect properties to real entities), or failed resolution against the catalog or an external knowledge base. Ji et al. [74] provide a comprehensive taxonomy distinguishing between factuality hallucinations (contradicting established knowledge) and faithfulness hallucinations (inconsistency with provided context). For retrieval-augmented generation (RAG) systems, counterfactual robustness testing can be applied: evaluating whether the model appropriately handles retrieved documents containing incorrect or contradictory information, or measuring whether outputs change when relevant documents are replaced with irrelevant ones [44]. High-quality RAG systems should show substantial sensitivity to document relevance—changing outputs significantly when retrieved context changes—while maintaining the ability to reject answers when retrieved documents lack relevant information. In MRS, this can be quantified as the proportion of recommended tracks that do not exist in the catalog or have incorrectly attributed metadata (artist, album, genre).
2) **Popularity, temporal and language bias**: This dimension evaluates systematic disparities in exposure across stratified groups, such as mainstream versus long-tail content or Western versus non-Western music. Abdollahpouri et al. [2] introduce the Average Recommendation Popularity (ARP) metric: $\text{ARP} = \frac{1}{|U|} \sum_{u \in U} \frac{\sum_{i \in R_u} \phi(i)}{|R_u|}$, where $\phi(i)$ represents item i's popularity score and R_u is the recommendation list for user u. They demonstrate that collaborative filtering algorithms exhibit systematic bias toward popular items, with ARP values significantly exceeding catalog median popularity. LLMs may exacerbate this bias or introduce new types of bias. Deldjoo et al. [30] demonstrate that ChatGPT-based recommender systems exhibit provider fairness issues, temporal instability, and recency bias in music and movie recommendations, evaluating consistency through repeated queries and temporal drift analysis. Sguerra et al. [144] reveal that LLM-generated musical taste profiles exhibit systematic quality variations across user mainstreamness and cultural diversity, measuring profile accuracy through genre precision and user study evaluations.
3) **Profile hazards**: This dimension assesses the system's sensitivity to variations in user profile representation, including robustness to profile edits (addition/removal of preference signals), degradation with truncated profiles (measuring utility loss when profile length is constrained), and the impact of noisy or contradictory preference signals. Wang et al. [161] introduce profile defect diagnostics, categorizing issues as inaccurate (contradictory preferences), incomplete (missing critical attributes), or both. Evaluation involves measuring recommendation stability (via Rank-Biased Overlap [168]) across profile perturbations and quantifying the accuracy-length trade-off: plotting recommendation quality (nDCG, recall@k) against profile token length to identify optimal compression points. For LLM-based systems, this includes assessing whether prompt length constraints (e.g., 2048 vs. 8192 tokens) significantly degrade personalization quality.
4) **Evaluator bias**: When LLMs serve as automatic evaluators (LLM-as-judge), these metrics quantify systematic divergence between LLM-based assessments and human ground-truth judgments. Zheng et al. [191] show that LLM judges exhibit multiple forms of bias: position bias (where GPT-4 shows 65% consistency when answer positions are swapped, favoring the first position in 30% of cases), verbosity bias (where judges may prefer longer responses, with GPT-4 showing 8.7% susceptibility to verbosity attacks compared to over 90% for weaker models), and self-enhancement bias (where GPT-4 exhibits approximately 10% higher win rates for its own outputs compared to human judgments). To measure evaluator bias, they propose conducting position swap tests